

Gaohan Gao

Undergraduate Student in Software Engineering | Xi'an Jiaotong University
Xi'an, Shaanxi, China | gaohan050121@outlook.com | github.com/gaohan0121

RESEARCH PROFILE

Undergraduate researcher working on multimodal artificial intelligence, remote sensing foundation models, vision-language learning, and agentic AI. Current work studies multimodal large language models for hyperspectral image change detection, SAR-RGB cross-modal retrieval, and tool-augmented multi-agent systems.

EDUCATION

Xi'an Jiaotong University | Xi'an, China

B.E. in Software Engineering | September 2023 - Present

RESEARCH EXPERIENCE

Multimodal Large Language Models for Hyperspectral Image Change Detection | January 2026 - Present

- Reformulated pixel-level change detection as a vision-language question-answering task for semantic interpretation.
- Refined ground-truth labels and designed an automated pipeline for region-level captions and question-answer pairs.
- Adapted multimodal models with supervised fine-tuning and parameter-efficient fine-tuning; conducted evaluation and analysis.

Structure-Aware SAR-RGB Cross-modal Image Retrieval | September 2025 - December 2025

- Investigated cross-modal retrieval across heterogeneous sensing modalities using CLIP-style feature alignment.
- Implemented strip-convolution modules to capture elongated ship structures and evaluated retrieval performance.
- Completed model training, hyperparameter tuning, ablation studies, and comparative analysis.

Deep Learning for Cytological Diagnosis of Pleomorphic Adenoma | September 2025 - April 2026

- Contributed to literature review, technical planning, data preparation, model discussion, and experimental analysis.

PUBLICATIONS

HSICD: Multimodal Large Language Model for Hyperspectral Image Change Detection. Gaohan Gao et al. *IEEE Geoscience and Remote Sensing Magazine*. Under review.

A Unified Algorithmic Framework for Battery Life Prediction Based on CTMC and Monte Carlo Simulation. Gaohan Gao et al. *MEEML 2026*. Co-first author.

SELECTED PROJECT

Multi-Agent Literature Research Assistant | Independent project, March 2026

- Designed a multi-agent workflow for paper discovery, ranking, question answering, novelty analysis, and literature synthesis.
- Integrated tool-augmented agents with Zotero and Obsidian for research knowledge-base synchronization.
- Built agent tracing for observable coordination and tool-use workflows using MCP, A2A, and RAG concepts.

TECHNICAL SKILLS

Programming: Python, C++, Java

Artificial Intelligence: PyTorch, Transformers, CLIP, LoRA, LLM fine-tuning

Research Areas: Multimodal LLMs, remote sensing, vision-language learning, change detection, cross-modal retrieval, agentic AI